



# Configuring an SDN Controller in Open Source Mininet Emulator

The exponential increase in the use of the Internet, coupled with the growth in cloud computing, has created a need for better network control and management. Software defined networking (SDN) – a means to control the network through software – is a solution to the problem. This tutorial, aimed at researchers working in the SDN and the cloud computing domain, presumes knowledge of Mininet Emulator, Python and SDN and would be of interest to B. Tech, M. Tech and PhD scholars looking for an open source simulation platform for SDN.

**S**DN is a revolutionary idea in computer networking that ensures significant flexibility and simplicity in network control and management, apart from giving a broader scope for innovation through programmability. SDN is all about controlling the network through software or, in other words, making the network programmable.

There has been a dramatic increase in Internet usage over the past few years, as compared to the use of networking technology, which hasn't been engaged as much. SDN has been introduced as a replacement for conventional networking, to meet market requirements. Fortunately, SDN couldn't have evolved at a better time. Since the use of cloud computing is increasing, it is necessary to automate the configurations as much as possible.

According to the ONF (Open Network Foundation) definition, SDN is actually a decoupling of the network control plane and the forwarding plane. With the demand

for cloud computing increasing, SDN has evolved as the most efficient way of controlling networking using some high level language to make the programming as flexible as possible.

## Traditional networking versus SDN

The main difference between traditional networking and SDN is the way in which data is handled and forwarded. Unlike traditional networking, SDN has separate devices called SDN controllers which control the data path. These control the path of packets coming to switches known as the OFvSwitch (Open Flow Virtual Switch). So, there has to be some interface between the forwarding plane and the controlling plane, which is provided by the OF (Open Flow) protocol. SDN enables admins to control the way switches handle the data, provide QoS (Quality of Service), and automate the process to make it less tedious and erroneous.